Containerization_and_DevOps

Experiment 3:- Docker Deploy NGINX Using Different Base Images and Comparing Image Layers

Part 1: Deploy NGINX Using Official Image (Recommended Approach)

Step 1: Pull the Image

```
docker pull nginx:latest
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ docker pull nginx:latest
latest: Pulling from library/nginx
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$
```

Step 2: Run the Container

```
docker run -d --name nginx-official -p 8080:80 nginx
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ docker run -d --name nginx-official -p 8080:80 nginx
2c2a5d3d1bfe39a727e3dd8a677d925c92bd1889cbe9ac6a2f55c04b9d097ebe
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$
```

Step 3: Verify

```
curl http://localhost:8080
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$
```

You should see the **NGINX welcome page**.

Key Observations

```
docker images nginx
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ docker images nginx
REPOSITORY   TAG       IMAGE ID       CREATED        SIZE
nginx        latest    341bf0f3ce6c   2 weeks ago    240MB
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$
```

- Image is **pre-optimized**
- Minimal configuration required
- Uses Debian-based OS internally

Part 2: Custom NGINX Using Ubuntu Base Image

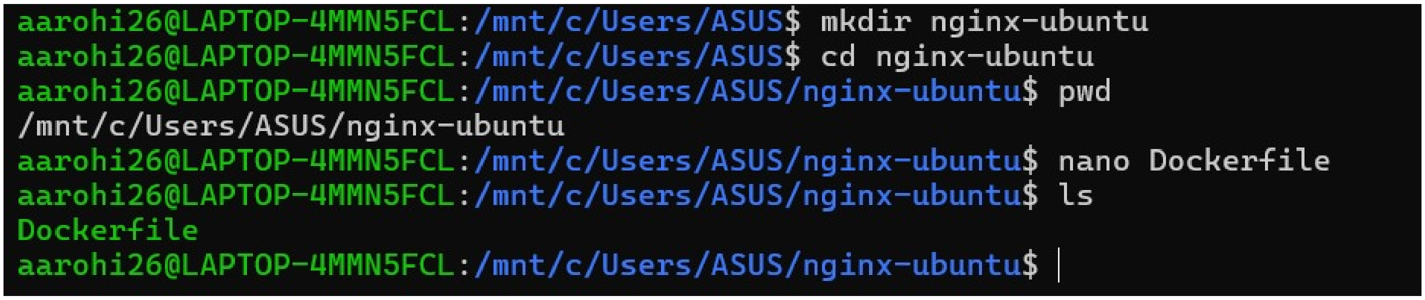
Step 1: Create Dockerfile

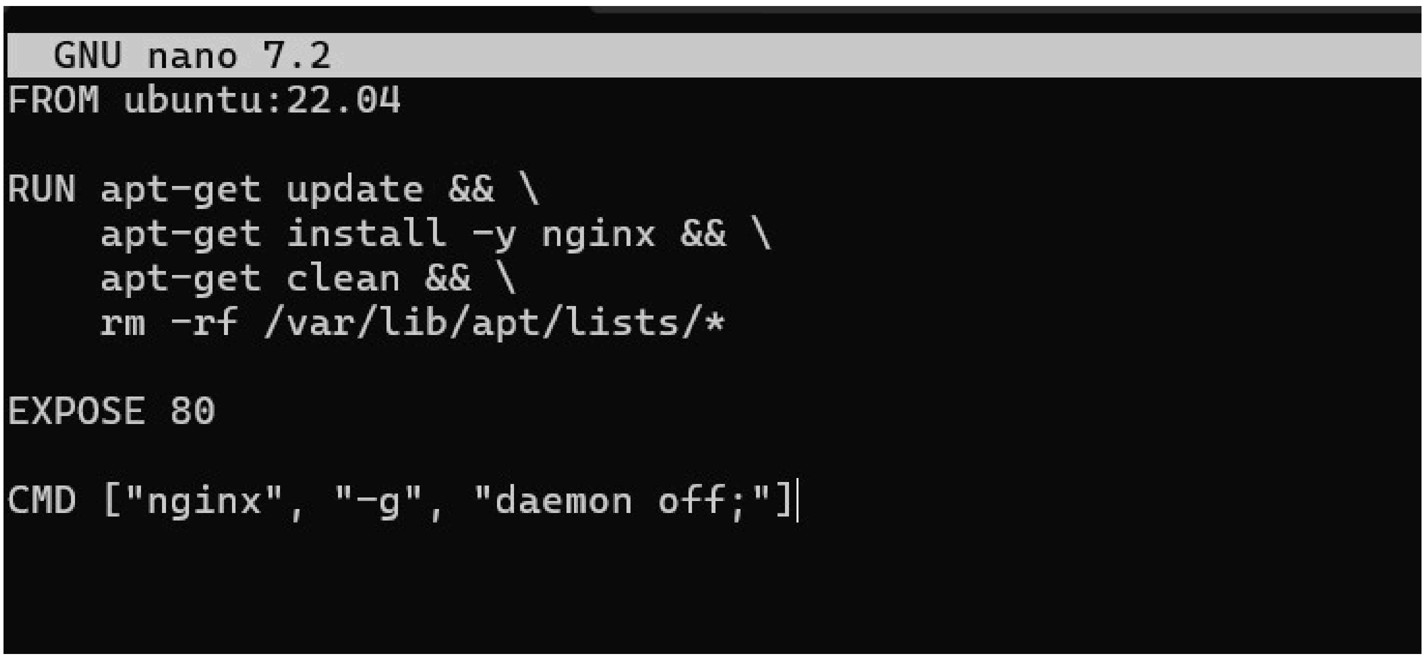
```
FROM ubuntu:22.04
```

```
RUN apt-get update && \
    apt-get install -y nginx && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```





Step 2: Build Image

```
docker build -t nginx-ubuntu .
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-ubuntu$ docker build -t nginx-ubuntu .
[+] Building 90.0s (7/7) FINISHED                                                                              docker:default
 => [internal] load build definition from Dockerfile                                                                     0.1s
 => => transferring dockerfile: 218B                                                                                     0.1s
 => [internal] load metadata for docker.io/library/ubuntu:22.04                                                          9.3s
 => [auth] library/ubuntu:pull token for registry-1.docker.io                                                            0.0s
 => [internal] load .dockerignore                                                                                        0.0s
 => => transferring context: 2B                                                                                          0.0s
 => [1/2] FROM docker.io/library/ubuntu:22.04@sha256:3ba65aa20f86a0fad9df2b2c259c613df006b2e6d0bfcc8a146afb8c525a9751    12.3s
 => => resolve docker.io/library/ubuntu:22.04@sha256:3ba65aa20f86a0fad9df2b2c259c613df006b2e6d0bfcc8a146afb8c525a9751     0.0s
 => => sha256:b1cba2e842ca52b95817f958faf99734080c78e92e43ce609cde9244867b49ed 29.54MB / 29.54MB                         11.3s
 => => extracting sha256:b1cba2e842ca52b95817f958faf99734080c78e92e43ce609cde9244867b49ed                                 1.0s
 => [2/2] RUN apt-get update &&     apt-get install -y nginx &&     apt-get clean &&     rm -rf /var/lib/apt/lists/*    64.9s
 => exporting to image                                                                                                   3.1s
 => => exporting layers                                                                                                  2.4s
 => => exporting manifest sha256:94d5856ec8a82c0230f62ec767cf2cf301fe48982f170926e29563b3fb3eaf02                        0.0s
 => => exporting config sha256:c97ef0abbcc3408b7d195812fd728121d2ef5929efa7924a0f63d0a20432bb2c                          0.0s
 => => exporting attestation manifest sha256:b43cb80a09b5dbe06553c7b425b216b4c84b89a14278851fc76c754ff2233da1            0.0s
 => => exporting manifest list sha256:5fb9b602c9b178a136c69b470557fb86331a4fc48dff2c27126bd20344cbad52                   0.0s
 => => naming to docker.io/library/nginx-ubuntu:latest                                                                   0.0s
 => => unpacking to docker.io/library/nginx-ubuntu:latest                                                                0.5s
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-ubuntu$
```

Step 3: Run Container

```
docker run -d --name nginx-ubuntu -p 8081:80 nginx-ubuntu
```/
![Run Container](/Containerization_and_DevOps/Labwork/Experiment3/Images/Img8.png)

Observations

```bash
docker images nginx-ubuntu
```

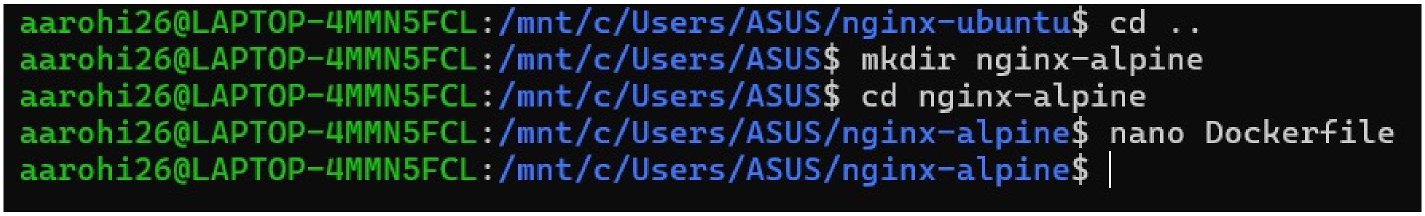
```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-ubuntu$ docker images nginx-ubuntu
REPOSITORY     TAG       IMAGE ID       CREATED              SIZE
nginx-ubuntu   latest    5fb9b602c9b1   About a minute ago   199MB
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-ubuntu$
```

- Much **larger image size**
- More layers
- Full OS utilities available

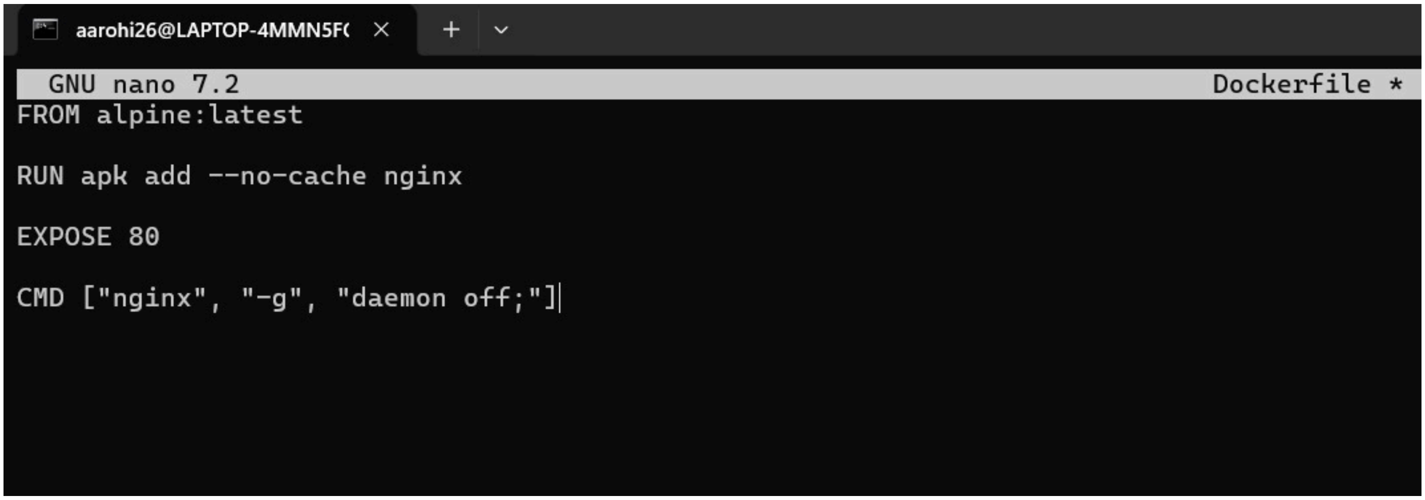
Part 3: Custom NGINX Using Alpine Base Image

Step 1: Create Dockerfile

```
FROM alpine:latest

RUN apk add --no-cache nginx

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```



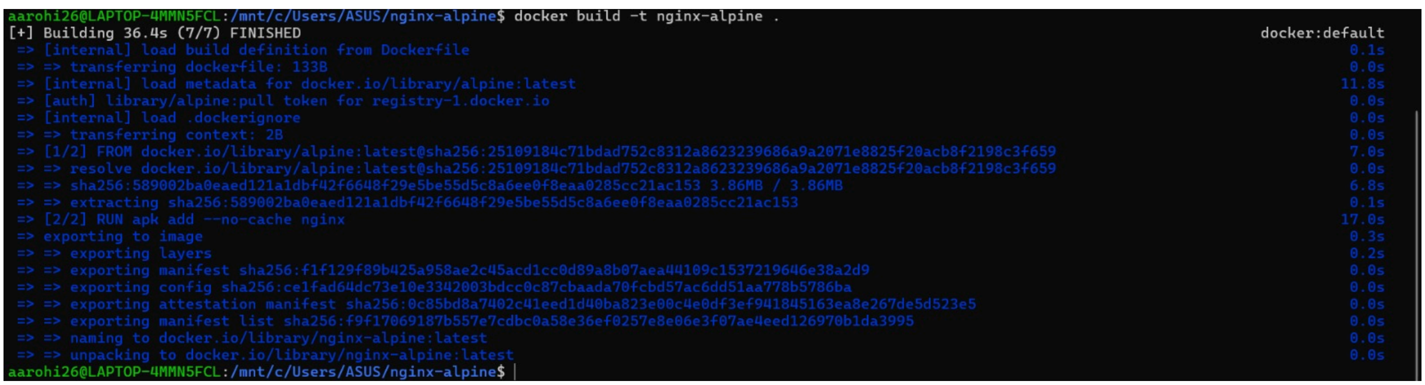
```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-ubuntu$ cd ..
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ mkdir nginx-alpine
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ cd nginx-alpine
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$ nano Dockerfile
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$
```



```
aarohi26@LAPTOP-4MMN5F(
  GNU nano 7.2                                                    Dockerfile *
FROM alpine:latest

RUN apk add --no-cache nginx

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Step 2: Build Image

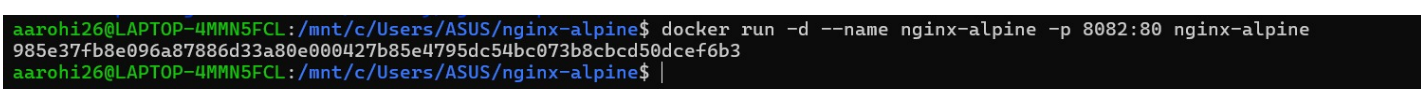
```
docker build -t nginx-alpine .
```



```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$ docker build -t nginx-alpine .
[+] Building 36.4s (7/7) FINISHED                                                                      docker:default
 => [internal] load build definition from Dockerfile                                                              0.1s
 => => transferring dockerfile: 133B                                                                              0.0s
 => [internal] load metadata for docker.io/library/alpine:latest                                                 11.8s
 => [auth] library/alpine:pull token for registry-1.docker.io                                                     0.0s
 => [internal] load .dockerignore                                                                                 0.0s
 => => transferring context: 2B                                                                                   0.0s
 => [1/2] FROM docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659   7.0s
 => => resolve docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659   0.0s
 => => sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153 3.86MB / 3.86MB                    6.8s
 => => extracting sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153                         0.1s
 => [2/2] RUN apk add --no-cache nginx                                                                           17.0s
 => exporting to image                                                                                            0.3s
 => => exporting layers                                                                                           0.2s
 => => exporting manifest sha256:f1f129f89b425a958ae2c45acd1cc0d89a8b07aea44109c1537219646e38a2d9                 0.0s
 => => exporting config sha256:ce1fad64dc73e10e3342003bdcc0c87cbaada70fcbd57ac6dd51aa778b5786ba                   0.0s
 => => exporting attestation manifest sha256:0c85bd8a7402c41eed1d40ba823e00c4e0df3ef941845163ea8e267de5d523e5     0.0s
 => => exporting manifest list sha256:f9f17069187b557e7cdbc0a58e36ef0257e8e06e3f07ae4eed126970b1da3995            0.0s
 => => naming to docker.io/library/nginx-alpine:latest                                                            0.0s
 => => unpacking to docker.io/library/nginx-alpine:latest                                                         0.0s
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$
```

Step 3: Run Container

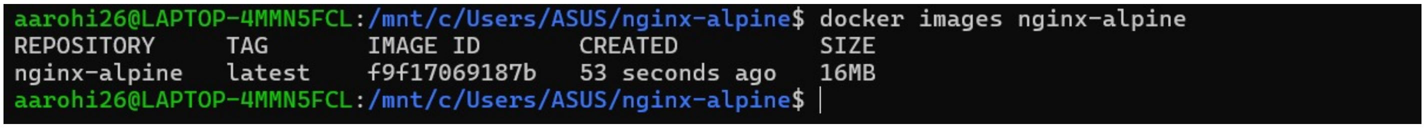
```
docker run -d --name nginx-alpine -p 8082:80 nginx-alpine
```



```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$ docker run -d --name nginx-alpine -p 8082:80 nginx-alpine
985e37fb8e096a87886d33a80e000427b85e4795dc54bc073b8cbcd50dcef6b3
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$
```

Observations

```
docker images nginx-alpine
```



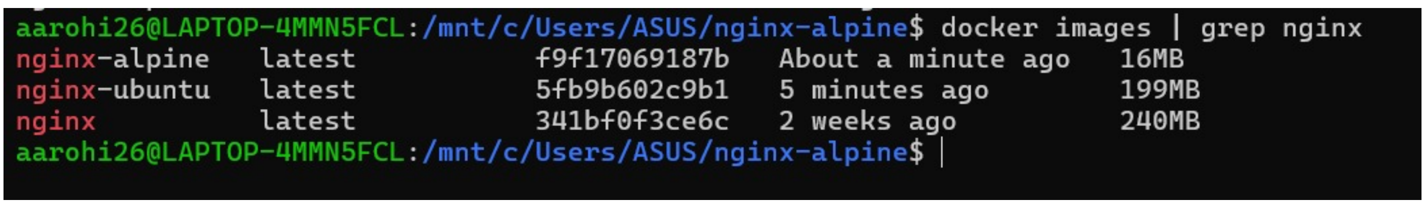
```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$ docker images nginx-alpine
REPOSITORY     TAG       IMAGE ID       CREATED          SIZE
nginx-alpine   latest    f9f17069187b   53 seconds ago   16MB
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$
```

- Extremely **small image**
- Fewer packages
- Faster pull and startup time

Part 4: Image Size and Layer Comparison

Compare Sizes

```
docker images | grep nginx
```



```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$ docker images | grep nginx
nginx-alpine   latest          f9f17069187b   About a minute ago   16MB
nginx-ubuntu   latest          5fb9b602c9b1   5 minutes ago        199MB
nginx          latest          341bf0f3ce6c   2 weeks ago          240MB
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$
```

Typical result (approx):

Image Type	Size
nginx:latest	240 MB
nginx-ubuntu	199 MB
nginx-alpine	16 MB

Inspect Layers

```
docker history nginx
docker history nginx-ubuntu
docker history nginx-alpine
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$ docker history nginx
docker history nginx-ubuntu
docker history nginx-alpine
IMAGE          CREATED       CREATED BY                                      SIZE      COMMENT
341bf0f3ce6c   2 weeks ago   CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   STOPSIGNAL SIGQUIT                              0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   EXPOSE map[80/tcp:{}]                           0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENTRYPOINT ["/docker-entrypoint.sh"]            0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   COPY 30-tune-worker-processes.sh /docker-ent…   16.4kB    buildkit.dockerfile.v0
<missing>      2 weeks ago   COPY 20-envsubst-on-templates.sh /docker-ent…   12.3kB    buildkit.dockerfile.v0
<missing>      2 weeks ago   COPY 15-local-resolvers.envsh /docker-entryp…   12.3kB    buildkit.dockerfile.v0
<missing>      2 weeks ago   COPY 10-listen-on-ipv6-by-default.sh /docker…   12.3kB    buildkit.dockerfile.v0
<missing>      2 weeks ago   COPY docker-entrypoint.sh / # buildkit          8.19kB    buildkit.dockerfile.v0
<missing>      2 weeks ago   RUN /bin/sh -c set -x     && groupadd --syst…   86.6MB    buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV DYNPKG_RELEASE=1~trixie                     0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV PKG_RELEASE=1~trixie                        0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV ACME_VERSION=0.3.1                          0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV NJS_RELEASE=1~trixie                        0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV NJS_VERSION=0.9.5                           0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV NGINX_VERSION=1.29.5                        0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   LABEL maintainer=NGINX Docker Maintainers <d…   0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   # debian.sh --arch 'amd64' out/ 'trixie' '@1…   87.4MB    debuerreotype 0.17
IMAGE          CREATED         CREATED BY                                      SIZE      COMMENT
5fb9b602c9b1   6 minutes ago   CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      6 minutes ago   EXPOSE [80/tcp]                                 0B        buildkit.dockerfile.v0
<missing>      6 minutes ago   RUN /bin/sh -c apt-get update &&     apt-get…   58.6MB    buildkit.dockerfile.v0
<missing>      9 days ago      /bin/sh -c #(nop)  CMD ["/bin/bash"]            0B
<missing>      9 days ago      /bin/sh -c #(nop) ADD file:52c0e467fa2e92f10…   87.5MB
<missing>      9 days ago      /bin/sh -c #(nop)  LABEL org.opencontainers.…   0B
<missing>      9 days ago      /bin/sh -c #(nop)  LABEL org.opencontainers.…   0B
<missing>      9 days ago      /bin/sh -c #(nop)  ARG LAUNCHPAD_BUILD_ARCH     0B
<missing>      9 days ago      /bin/sh -c #(nop)  ARG RELEASE                  0B
IMAGE          CREATED         CREATED BY                                      SIZE      COMMENT
f9f17069187b   2 minutes ago   CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      2 minutes ago   EXPOSE [80/tcp]                                 0B        buildkit.dockerfile.v0
<missing>      2 minutes ago   RUN /bin/sh -c apk add --no-cache nginx # bu…   2.2MB     buildkit.dockerfile.v0
<missing>      3 weeks ago     CMD ["/bin/sh"]                                 0B        buildkit.dockerfile.v0
<missing>      3 weeks ago     ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /…   9.11MB    buildkit.dockerfile.v0
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$ |
```

Observations:

- Ubuntu has many filesystem layers
- Alpine has minimal layers
- Official NGINX image is optimized but heavier than Alpine

Part 5: Functional Tasks Using NGINX

Task 1: Serve Custom HTML Page

```
mkdir html
echo "<h1>Hello from Docker NGINX</h1>" > html/index.html
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$ mkdir html
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine$ cd html
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine/html$ nano index.html
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/nginx-alpine/html$ |
```

```
aarohi26@LAPTOP-4MMN5F(
  GNU nano 7.2                                                 index.html *
<h1>Hello from Docker NGINX</h1>
```

Run:

```
docker run -d \
  -p 8083:80 \
  -v $(pwd)/html:/usr/share/nginx/html \
  nginx
```

```
aarohi26@LAPTOP-4MMN5FCL:~$ docker run -d \
  -p 8083:80 \
  -v $(pwd)/html:/usr/share/nginx/html \
  nginx
09eb8b8716b5891dc9333fc143cde50dc7e4104a829dc685bcaa99341a1a4368
aarohi26@LAPTOP-4MMN5FCL:~$ |
```

Task 2: Reverse Proxy (Conceptual)

NGINX can:

- Forward traffic to backend services
- Load balance multiple containers
- Terminate SSL

Example use cases:

- Frontend for microservices
- API gateway
- Static file server

Part 6: Comparison Summary

Image Comparison Table

Feature	Official NGINX	Ubuntu + NGINX	Alpine + NGINX
Image Size	Medium	Large	Very Small
Ease of Use	Very Easy	Medium	Medium
Startup Time	Fast	Slow	Very Fast
Debugging Tools	Limited	Excellent	Minimal
Security Surface	Medium	Large	Small
Production Ready	Yes	Rarely	Yes

Part 7: When to Use What

Use Official NGINX Image When:

- Production deployment
- Standard web hosting
- Reverse proxy / load balancer

Use Ubuntu-Based Image When:

- Learning Linux + NGINX internals
- Heavy debugging needed
- Custom system-level dependencies

Use Alpine-Based Image When:

- Microservices
- CI/CD pipelines
- Cloud and Kubernetes workloads